

Operator-Aware World Model Compression

Preserving planning behavior, not just prediction

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Motivation

- ▶ World models are increasingly used for planning and control.
- ▶ Standard compression methods preserve weights, activations, or prediction error.
- ▶ But in planning, the model is used to choose actions.

Problem

Standard compression question

Can we make the model smaller while preserving prediction accuracy?

Our question

Can we make the model smaller while preserving the induced planning operator?

Background: Latent World-Model Planning

$$h_t, g, A_{1:H}^{(1)}, \dots, A_{1:H}^{(N)} \longrightarrow \text{world model} \longrightarrow c_1, \dots, c_N$$

$$A^* = \arg \min_i c_i$$

Planning pipeline diagram

Models Studied

LeWM

- ▶ Latent world model from pixels.
- ▶ ViT-style encoder.
- ▶ Learned predictor / transition model.
- ▶ Used with CEM/MPC planning.

DINO-WM / SWM-PreJEPA

- ▶ Frozen visual encoder.
- ▶ Learned Transformer predictor.
- ▶ Action/proprio conditioning.
- ▶ Same planning-style deployment.

Key Idea: Compress the Planning Operator

$$\mathcal{O}_\theta(h_t, g, A_{1:H}) = C_\theta(A_{1:H})$$

- ▶ The model induces a cost over candidate action sequences.
- ▶ CEM/MPC only needs the cost ranking to select actions.
- ▶ A compressed model may preserve planning even if latent predictions are imperfect.

What Does Operator Preservation Mean?

- ▶ **Rank correlation:** do teacher and student rank actions similarly?
- ▶ **Top- k elite overlap:** do they agree on the CEM elite set?
- ▶ **Teacher regret:** how bad is the student's chosen action under the teacher?
- ▶ **First-action error:** does MPC choose a similar first action?

Why Prediction MSE Does Not Preserve Planning

Planning is a ranking problem

The teacher selects

$$A_T^* = \arg \min_A C_T(A).$$

If the teacher action gap is

$$\Delta = \min_{A \neq A_T^*} (C_T(A) - C_T(A_T^*)),$$

then the student preserves the teacher action only if

$$\sup_A |C_S(A) - C_T(A)| < \frac{\Delta}{2}.$$

- ▶ Prediction MSE gives an average latent error, not a uniform cost-ranking guarantee.
- ▶ When two candidate actions have close costs, small prediction errors can flip the

Compression Setup

Teacher

- ▶ Full LeWM or PreJEPA world model.
- ▶ Generates action costs and elite sets.

Student

- ▶ Compressed predictor / transition operator.
- ▶ Encoder fixed initially.
- ▶ Trained or calibrated to preserve planning behavior.

Compression Methods

- 1 Weight-SVD predictor compression.
- 2 Activation-aware SVD.
- 3 Prediction-only distillation.
- 4 Operator-aware distillation.

Operator-Aware Loss

Teacher and student costs over the same candidate actions:

$$c_T^{(i)} = \mathcal{O}_{\theta_T}(h_t, g, A_{1:H}^{(i)}), \quad c_S^{(i)} = \mathcal{O}_{\theta_S}(h_t, g, A_{1:H}^{(i)}).$$

Soft cost distribution:

$$p_T(i) = \frac{\exp(-c_T^{(i)} / \tau)}{\sum_j \exp(-c_T^{(j)} / \tau)}, \quad p_S(i) = \frac{\exp(-c_S^{(i)} / \tau)}{\sum_j \exp(-c_S^{(j)} / \tau)}.$$

$$\mathcal{L}_{\text{op}} = D_{\text{KL}}(p_T \parallel p_S) + \lambda \mathcal{L}_{\text{elite}}.$$

Operator Cache

- ▶ Sample state/goal pairs from the offline dataset.
- ▶ Sample candidate action sequences.
- ▶ Store teacher costs, top- k indices, elite sets, and selected actions.
- ▶ Reuse the same cache across compression methods.

Operator cache schema

Experimental Protocol

Environments

- ▶ TwoRoom
- ▶ PushT
- ▶ Cube

Models

- ▶ LeWM
- ▶ SWM-PreJEPA

Metrics

- ▶ Success rate
- ▶ Top- k overlap
- ▶ Regret
- ▶ Model size

Main Result: Compression Frontier

Compression ratio vs. planning success

Diagnostics: Does Prediction Error Explain Planning?

Latent MSE vs. success

*Top- k overlap / regret vs.
success*

Limitations

- ▶ Operator cache depends on sampled states, goals, and action candidates.
- ▶ Preserving teacher behavior does not guarantee optimal behavior.
- ▶ Initial compression targets the predictor, not the visual encoder.
- ▶ Full benchmark cost depends on CEM settings and environment difficulty.

Conclusion

Main idea

For world models used in planning, compression should preserve the induced action-ranking operator.

- ▶ Standard compression asks whether the model still predicts.
- ▶ We ask whether the compressed model still chooses useful actions.
- ▶ Operator-aware compression targets rank, elite-set, and MPC-action preservation.

Q & A

Questions?